

Supplementary Information for “Geometric frustration induces the transition between rotation and counterrotation in swirled granular media”

VIDEO

Included:

Experimental lab frame: rotating, stall, counterrotating
 Experimental M-frame: rotating, stall, counterrotating
 Numerical lab frame all friction: rotating, counterrotating
 Numerical M-frame all friction: rotating, counterrotating
 Numerical lab frame no disc friction: rotating, counterrotating
 Numerical M-frame no disc friction: rotating, counterrotating
 Numerical lab frame no wall friction: rotating, counterrotating
 Numerical M-frame no wall friction: rotating, counterrotating
 Minimal model M-frame: no slip and slip
 Minimal model lab frame: no slip and slip

CHANGING TO A ROTATING FRAME OF REFERENCE

In this section we give more details on how to convert between the lab frame and the M-frame. See also [3, Section 39].

Recall that the M-frame is the rotating frame of reference obtained by rotating with angular velocity ω about point S, the axis of swirling. Let the origin of the coordinate systems for both the lab and M-frames be momentarily at S. Consider a point particle at location x (which is momentarily the same in both frames), with velocities v_L, v_M in the lab and M-frames respectively. By the definition of angular velocity, these velocities are related by

$$v_L = v_M + \omega \times x. \quad (1)$$

Now consider a rigid body with center of mass at x moving with translational velocities v_L, v_M and rotating with angular velocities about its center of mass ω_L, ω_M in the lab and M-frames respectively. Let x' be some point on the body. Its velocity in the M-frame is $v'_M = v_M + \omega_M \times (x' - x)$, by the definition of angular velocity in the M-frame, and its velocity in the lab frame is $v'_L = v'_M + \omega \times x'$, by (1). Substituting for v'_M gives $v'_L = v_M + \omega_M \times (x' - x) + \omega \times x' = v_M + \omega \times x + (\omega_M + \omega) \times (x' - x)$. We also know that $v'_L = v_L + \omega_L \times (x' - x)$, by the definition of angular velocity in the lab frame, so the translational velocities are related by (1) and the angular velocities are related by

$$\omega_L = \omega + \omega_M. \quad (2)$$

In particular, this implies that a rigid body with:

- No rotation in the lab frame $\Rightarrow$ rotates at $-\omega$ in the M-frame;
- No rotation in the M-frame $\Rightarrow$ rotates at ω in the lab frame.

Therefore, the boundary of the container rotates at constant angular velocity $-\omega$ in the M-frame.

To see how the fictitious forces arise, let $\frac{d}{dt_L}, \frac{d}{dt_M}$ be the time-derivatives that an observer in the lab and M-frames would calculate respectively, and to rewrite (1) as [4]

$$\frac{dx}{dt_L} = \frac{dx}{dt_M} + \omega \times x \quad \Leftrightarrow \quad \frac{d}{dt_L} = \frac{d}{dt_M} + \omega \times. \quad (3)$$

Therefore

$$m \frac{d^2 x}{dt_L^2} = m \left(\frac{d}{dt_M} + \omega \times \right)^2 x = m \frac{d^2 x}{dt_M^2} + 2\omega \times \frac{dx}{dt_M} + \omega \times (\omega \times x). \quad (4)$$

The last two terms represent fictitious forces that a particle in a rotating frame of reference feels, due to the acceleration of the frame. That is, if there are no external forces on a particle in the lab frame, then there are two external forces on the particle in the M-frame:

$$\begin{aligned} \text{Coriolis force:} & \quad 2v_M \times \omega \\ \text{Centrifugal force:} & \quad \omega \times (x \times \omega) \end{aligned}$$

These forces are sketched in Figure 1c of the main text.

CALCULATION OF LOCATION THAT PARTICLES BECOME PINNED AND FREE (FIG 3)

Becoming pinned

Particles experience multiple collisions while transitioning from the loose to the pinned region. The angular location on the dish at which particles become pinned is therefore determined by the average angle at which these collisions occur. A collision event is assigned to a particle p at time t if the particle experiences a sufficiently sharp turn in its trajectory, calculated using a threshold on the second derivative of its coordinates $x(t, p)$ and $y(t, p)$.

Becoming free

The angular location on the dish at which particles become free is determined by the angular position at which the particle no longer travels parallel to the dish edge. To determine this, we calculate the best fit slope for the particle's coordinates between the preceding 7 frames and the following 7 frames. When this slope deviates from the perpendicular to the particle's radial vector on the dish (or the particle's azimuthal vector) by more than 10 degrees, we mark this particle as becoming "free." Mathematically:

$$\theta_{\text{free}} = \sum_{t=7}^{\text{all frames}-7} \sum_{p=1}^{\text{all particles}} \theta(t, p) * \text{boolean}(|\theta(t, p) - \arctan(-\frac{x(t, p) - x_{\text{dish center}}}{y(t, p) - y_{\text{dish center}}})| > 10),$$

where

$$\theta(t, p) = \frac{\sum_{i=t-7}^{t+7} (x(i, p) - \bar{X})(y(i, p) - \bar{Y})}{\sum_{i=t-7}^{t+7} (x(i, p) - \bar{X})^2},$$

and

$$\bar{X} = \frac{\sum_{i=t-7}^{t+7} x(i, p)}{15}, \bar{Y} = \frac{\sum_{i=t-7}^{t+7} y(i, p)}{15}.$$

CALCULATION OF PARTICLE FRUSTRATION (FIG 4B)

Particle Image Velocimetry (using MATLAB's PIVlab) is run on the particles within the pinned region. The PIV interrogation windows are set such that a total of 16 velocity vectors is calculated within each particle. The velocity values ($\vec{v}_{\text{PIV}}$) are then used to calculate the following:

Average velocity

Average velocity is the average of the velocity vector magnitudes:

$$\text{Average velocity} = \sum_{\text{all vectors}} |\vec{v}_{\text{PIV}}|.$$

Average particle spin

Average particle spin takes into account only the azimuthal velocity, calculated by averaging the dot product of each PIV vector with its azimuthal vector (the azimuthal vector of the coordinates at which the PIV arrows have been calculated, taking the middle of the particle as the origin):

$$\text{Average particle spin} = \sum_i^{\text{all vectors}} (\vec{v}_{\text{PIV},i} \cdot \frac{\text{azimuthal vector at point i}}{|\text{azimuthal vector at point i}|}).$$

NUMERICAL SIMULATION METHOD

Overview

Our simulations consider N identical discs of radius R_{disc} in a circular container of radius $R_{container}$. We treat the container as a disc with a different radius so the number of discs in the simulation is $N + 1$. The discs are labelled $0, 1, 2, \dots, N$, where 0 indexes the container. The discs have mass m and moment of inertia I . The container is assumed to have infinite mass and moment of inertia. The state of the system at any point in time is given by the positions of the discs and container $\{x_i\}_{i=0}^N$, $x_i \in \mathbb{R}^2$, the velocities of the discs and container $\{v_i\}_{i=0}^N$, $v_i \in \mathbb{R}^2$, and the angular velocities of the discs $\{\omega_i\}_{i=1}^N$ (the container does not rotate.)

The container is “swirled,” translated along a path which is topologically a loop, at frequency ω . We translate at constant speed along a path which is a regular polygon with M sides and amplitude A (distance from center of polygon to vertex.) This path approximates a circle but allows us to solve analytically for disc-container collisions.

The dynamics of the discs are assumed to satisfy Newton’s equations everywhere except at collisions, so the discs travel with constant linear and angular velocity. Notably, they do not experience friction with the floor. When discs collide with each other or the container, their linear and angular velocities are updated according to a “collision law,” a rule for updating velocities which is consistent with certain physical principles. We choose our disc-disc collisions to be perfectly elastic in the normal direction, and in the tangential direction apply an impulse consistent with kinetic Coulomb friction (we ignore static friction.) Our disc-container collisions are similar but we consider the container to have infinite mass and moment of inertia, so its velocity doesn’t change after a collision. More details on the collision law are given in section below.

The dynamics are implemented using an event-driven method. We consider three kinds of events:

1. Two discs collide;
2. A disc collides with the container;
3. The container reaches a vertex of its polygonal path.

The method proceeds by repeating the following steps.

- (i) Calculate the times of all future events;
- (ii) Determine the time of the event that happens the soonest;
- (iii) Determine the positions and velocities of the system at that time and then update them using the rule associated with the event.

When none of the events occurs we may solve analytically for the positions and velocities of all the discs. When an event occurs we must change the velocities using the rule associated with that event.

Finding the next event

In this section we describe how the times of future events are calculated, considering each type of event in turn.

1. We calculate the collision time t_{ij} for each pair of discs (i, j) . For some disc i with position x_i at the current time, the position at any time increment t from the current event (if there have been no collisions) is

$$p_i(t) = x_i + tv_i. \quad (5)$$

A pair of discs (i, j) in isolation will collide when the distance between their centers equals the sum of their radii:

$$|p_i(t) - p_j(t)|^2 = (2R_{disc})^2. \quad (6)$$

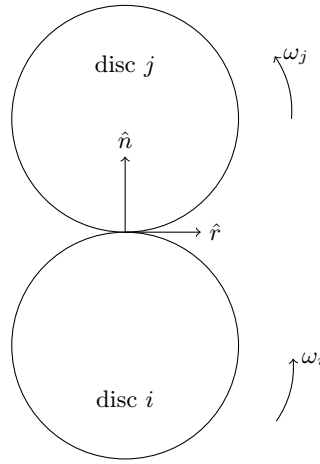


Figure 1. Sketch of two discs at the point of contact. Vectors $\hat{n}$, $\hat{r}$ are the normal and tangent directions as described in the text.

This is a quadratic equation in t with positive leading coefficient, whose roots give the collision times. If both roots are negative, then the discs will never collide on their present trajectories (though they may have collided in the past). If the roots have different signs, then the current distance between centers is less than $2R_{disc}$, and so the discs are overlapping; note that this should never happen if the simulation has been implemented correctly. If the roots are both positive, then the lesser of them is the time that the discs will collide. If the roots are not real, then the discs will never collide. Therefore we can set t_{ij} to be the lesser of the roots if it is real and positive, or $t_{ij} = \infty$ otherwise.

We make one small change to the above to account for numerical errors. It is sometimes the case that two discs that have just collided are predicted to collide next by the program (i.e., their solutions to (6) are real and positive.) In theory, the parabola in t from (6) should only have one root at the origin, but numerical errors may cause the roots to both be real and positive. When this has happened, the predicted next time of collision has always been less than 10^{-14} . To prevent these errors from being mistaken for actual collision predictions, we require that a predicted time of collision be greater than 10^{-13} , otherwise we ignore that prediction. The minimal time between collisions is around 10^{-11} and 10^{-9} with and without friction, respectively.

2. For collisions between discs and the container, the collision times are calculated in a similar way as they are for pairs of discs except now we use the quadratic equation

$$|p_i(t) - p_0(t)|^2 = |R_{container} - R_{disc}|^2 \quad (7)$$

and choose the greater of the roots. As long as the discs always stay within the container, this quadratic will always have real roots of different signs (except in the case that the container and the disc have exactly the same linear velocity.)

3. Calculating the time of the next update to the container's velocity is trivial, since the container reaches a vertex of its polygonal path at fixed time increments.

Our current method calculates the collision times of all pairs of discs, which is an $O(n^2)$ calculation. The method could be accelerated by using cell lists, making it an $O(n)$ calculation [1]. Our method was fast enough for the numbers of discs under consideration but simulating large numbers of discs would require cell lists.

Collision law

We now describe how to update the velocities of the discs after a collision. Our approach for relating tangential impulses to normal impulses is based on the calculations in [5], although our equations are simpler since we consider only discs, and [5] considers arbitrary rigid bodies. These calculations can be justified asymptotically if the time over which the collision takes place is much shorter than a typical evolution time [2].

Suppose discs i and j have just collided so their boundaries are exactly tangent, as in Figure 1. Let $\hat{n} = (x_j - x_i)/|x_j - x_i|$ be the unit vector normal to the point of contact, pointing from x_i to x_j , and let $\hat{r} = -(x_j - x_i)^\perp/|x_j - x_i|$ be the unit tangent vector at the contact. Here $(x, y)^\perp = (-y, x)$. For generality we let the discs have masses m_i, m_j and moments of inertia I_i, I_j .

Let $v_i, v_j, \omega_i, \omega_j$ be the velocities of the discs exactly when they contact but before the collision law has been implemented, and let $v'_i, v'_j, \omega'_i, \omega'_j$ be the velocities after the collision law. Let $v_{i(j)}^{norm}, v_{i(j)}^{tang}$ be the normal and tangential components of each disc's initial linear velocity, i.e. $v_i^{norm} = v_i \cdot \hat{n}$, $v_i^{tang} = v_i \cdot \hat{r}$, and similarly for v_j . The velocity components are updated by increments $\Delta v_{i(j)}^{norm}, \Delta v_{i(j)}^{tang}, \Delta \omega_i$ giving updated velocities

$$v'_i = (v_i^{norm} + \Delta v_i^{norm})\hat{n} + (v_i^{tang} + \Delta v_i^{tang})\hat{r}, \quad \omega'_i = \omega_i + \Delta \omega_i,$$

and similarly for v'_j .

The change in normal velocities is calculated assuming an elastic collision. Because we consider only discs, the normal force always points toward the centers of mass and does not mix with the tangential direction, so an elastic collision is equivalent to conserving energy and momentum conservation in the normal direction. (This is related to the equivalence of Newton's law of restitution, and the Poisson hypothesis, for discs, see [5].) The increments are

$$\begin{aligned} \Delta v_i^{norm} &= \frac{2m_j}{m_i + m_j}(v_j^{norm} - v_i^{norm}) \\ \Delta v_j^{norm} &= \frac{2m_i}{m_i + m_j}(v_i^{norm} - v_j^{norm}) = -\frac{m_i}{m_j}\Delta v_i^{norm}. \end{aligned}$$

We wish to derive tangential increments from the Coulomb friction law for kinetic friction. Suppose the collision is not instantaneous, but rather takes place over a time interval of length τ . Let $F_n(t), F_f(t)$ be the normal and tangential components of the force acting on disc j during the collision; $F_f(t)$ is the friction force. The forces acting on disc i are equal and opposite, by Newton's third law. The Coulomb friction law says that [3]

$$F_f(t) = s_0 \mu |F_n(t)|, \quad (8)$$

where μ is the coefficient of kinetic friction. The above holds provided the relative velocity at the point of contact is nonzero; after it hits zero $F_f(t) = 0$ and the relative velocity is zero for the remainder of the collision. The sign $s_0 \in \{-1, 1\}$ is chosen to ensure that friction reduces magnitude of the relative velocity of the points in contact. It is computed from the initial relative velocity v_{rel} (the velocity of the contact point on disc j minus the velocity of the contact point on disc i) as

$$s_0 = -\text{sgn}(v_{rel}), \quad v_{rel} = v_j^{tang} - v_i^{tang} + R_{disc}(\omega_i + \omega_j). \quad (9)$$

If we knew the normal force $F_n(t)$ and the length τ of the collision, then we could update the tangential velocities using Newton's second law by integrating (8). We don't know either $F_n(t)$ or τ (and they probably depend sensitively on the details of the collision), but, crucially, we know the total *impulse* in the normal direction, and this determines the total tangential impulse via (8). Indeed, following [5], we integrate (8) over the interval $[0, \tau]$ to obtain

$$J_f = s_0 \mu J_n, \quad (10)$$

where $J_f = \int_0^\tau F_f(t)dt$, $J_n = \int_0^\tau F_n(t)dt$ are the impulses in the tangential and normal directions (for disc-disc collisions, by the choice of axes we have $F_n(t) \geq 0$.) Equation (10) is valid provided the relative velocities never change sign; if they do the tangential impulse must be truncated as described below.

The impulse in the normal direction $\hat{n}$ for disc j is the change in the normal component of momentum,

$$J_n = m_j \Delta v_j^{norm}. \quad (11)$$

The magnitude of the maximum impulse in the tangential direction is therefore

$$J_f^{max} = \mu m_j \Delta v_j^{norm}. \quad (12)$$

The actual tangential impulse is

$$J_f = s_0 \min(J_f^{max}, J_f^{stick}), \quad (13)$$

where J_f^{stick} is the magnitude of the impulse required to make the relative tangential velocity between discs exactly zero (the superscript indicates the discs “stick” to each other in the tangential direction.) A formula for J_f^{stick} will be derived in (15) below.

Given J_f , the tangential increments are computed from the change in tangential and angular momenta:

$$m_j \Delta v_j^{tang} = J_f, \quad m_i \Delta v_i^{tang} = -J_f, \quad I_j \Delta \omega_j = R_{disc} J_f, \quad I_i \Delta \omega_i = R_{disc} J_f$$

The tangential and angular velocities are updated as

$$\begin{aligned} v_i'^{tang} &= v_i^{tang} - \frac{J_f}{m_i}, & \omega_i' &= \omega_i + \frac{R_{disc} J_f}{I_i} \\ v_j'^{tang} &= v_j^{tang} + \frac{J_f}{m_j}, & \omega_j' &= \omega_j + \frac{R_{disc} J_f}{I_j} \end{aligned} \quad (14)$$

The truncated impulse magnitude J_f^{stick} is the impulse required to make the relative tangential velocity between points in contact equal zero. It is found by solving equation

$$v_j^{tang} + \frac{J_f^{stick}}{m_j} - \left(v_i^{tang} - \frac{J_f^{stick}}{m_i} \right) + R_{disc} \left(\omega_j + \frac{R_{disc} J_f^{stick}}{I_j} + \omega_i + \frac{R_{disc} J_f^{stick}}{I_i} \right) = 0,$$

resulting in

$$J_f^{stick} = \frac{\left| v_j^{tang} - v_i^{tang} + R_{disc}(\omega_i + \omega_j) \right|}{\frac{1}{m_i} + \frac{1}{m_j} + \frac{R_{disc}^2}{I_i} + \frac{R_{disc}^2}{I_j}}. \quad (15)$$

Processing collisions with the container is similar, except that we set $m_0 = I_0 = \infty$. In this case, the initial increment is

$$\Delta v_i^{norm} = 2(v_j^{norm} - v_i^{norm})$$

$$\Delta v_j^{norm} = 0$$

Since $\omega_0 = 0$ we do not need to worry about the change in sign in some of the formulas because the collision is interior.

We make one final remark, that one can modify this method to allow for non-elastic behavior in the normal direction. However, it is then possible for discs to stick to each other, after which one has to track the dynamics of the cluster. Collision laws between clusters can become arbitrarily complicated as one solves for the propagation of impulse between the discs in each cluster. Since the dynamics we seek to understand do not seem to require nonelastic collisions we did not consider adding this additional complexity.

SIMULATION PARAMETERS

The container’s path is a polygon with $M = 30$ sides, and the period of the swirl is 11.866 units of time. The container moves at one unit of distance per unit of time, so the radius of the polygon formed by the container’s path is 1.892. Since the container’s radius is 8.6, this makes the ratio of the radius of the container’s path to the radius of the container equal to 0.22, which matches the experimental setup.

The following are the default parameters used in the plots and videos.

Number of discs	N	variable; typically 28-48
Mass of disc	m	1
Moment of inertia of disc	I	10
Radius of disc	R_{disc}	1
Radius of container	$R_{container}$	8.6
Sides in container’s polygonal path	M	30
Period of swirl	ω^{-1}	11.866071
Number of events calculated		10^6
Disc/Disc Friction Parameter	μ_d	1
Disc/Container Friction Parameter	μ_w	1

To remove the effect of a certain kind of friction from the simulation, we set that friction parameter to 0. A “low” friction value means it is set to 0.1, a “medium” friction value means it is set to 0.5, and a “high” friction value means it is set to 1.0. Note that setting both parameters to 0 will result in a crash as the discs’ energies will blow up.

Figure 5b in the main text was generated by taking the average over all events, for each N between 5 and 48, of the total angular velocity of the discs about their center of mass over 10^6 iterations.

For Figure 5c, the M-frame was divided evenly into a 100x100 grid of cells, and the amount of times that a disc’s center was found inside a cell was measured for $N = 28$ and 48. Cells that are colored red were found to contain disc centers very often.

For Figure 5d, the M-frame was divided evenly into a 20x20 grid of cells, and the average linear velocity of discs found in each of the cells was calculated for $N = 28$ and 48.

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